

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 412 — Past Questions & Materials
by Renance

Course: MTH 412
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — MTH 412 Functional Analysis 2018 2019 First Semester

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2018/2019

Course Code/Title: MTH 412 — Functional Analysis

Instructions: Attempt any 4 questions. Time Allowed: 2/2 Hours.

Question 1 (17% Marks)

- Define a linear map
 - When is a linear map said to be
 - bounded
 - continuous at a point
 - continuous everywhere.
 - Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a functional defined by $f(x) = x \cdot a$ where a is fixed and $X = (x_1, x_2, x_3)$ and $a = (a_1, a_2, a_3)$. Prove that f is (1) linear (11) bounded (9marks) (2'-marks)
- (6 marks)

Question 2 (17 'Z Marks)

- What is a dual space of a norm vector space X ? (2'marks)
- Suppose $(x_1, x_2, x_3, \dots, x_p)$ is a finite basis of a norm vector space X , prove that the finite combination of dual basis of X is linearly independent. (7marks)
- Let P be the vector space of polynomials in t over $\mathbb{R}$ defined by $P = \{a + bt : a, b \in \mathbb{R}\}$. Let $f, g: P \rightarrow \mathbb{R}$ be defined by $f(x(t)) = \int_0^1 x(t) dt$, $g(x(t)) = \int_0^2 x(t) dt$. Find the basis of f, g . (8marks)

Question 3 (17 4% Marks)

- What do you understand by a complete Inner product space? (2'Amarks)
- State and prove the Parallelogram equality theorem. (0marks)
- Define canonical imbedding mapping and prove that it is linear (6marks)

Question 4 (17% Marks)

- What is a Banach space? (2'smarks)
- Show that $\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$ is a Cauchy sequence (6marks)

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- Prove that $C[0,1]$ is not-complete under the integral norm for which $\|f\| = \int_0^1 |f(t)| dt$ (0marks)

Question 5 (17 2 Marks)

- Define the term norm vector space (3amarks)
- List 5 examples of a norm vector space (5marks)
- Given $f(x) = e^x$ on $[0,8]$. Prove that (1) $f(x) \in L[0,8]$ (2) $f(x) \in L^3[0,8]$ (9marks)

Question 6 (17 '2 Marks)

- a. Define a vector space over a field K (3emarks)
- b. Prove that the plane $\mathbb{R}^2$ is a vector space 7 (Omarks)
- c, Justify that the set of integers is not a vector space (Smarks)

Paper 2 of 2 — MTH412 Functional Analysis 2020 2021

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

: | DEPARTMENT OF MATHEMATICS

ar a FIRST SEMESTER EXAMINATION 2020/2021

oa ley Code/Title: MTH 412 — Functional Analysis

ions: Answer any 4 questions. Time Allowed: 2!4 Hours. NOTE: 30 Marks for C.A

Question 1

a. What is a linear functional. (2marks)

b. Let $X = 1, \dots$. For each $\mathbb{N} = (X_1, X_2, X_3, \dots)$ in $\mathbb{N}$, define $Tx = (0, \dots)$.

Prove that T is (i) a linear map on $\mathbb{N}$, (ii) bounded (iii) there exist $\|T\|$. (1 0marks)

c. Let $X = Y = \mathbb{R}$ be the real line. Define for each $x \in \mathbb{R}, T: \mathbb{R} \rightarrow \mathbb{R}$ by $Tx = 9x + 3$.

Prove that T is not a linear functional on $\mathbb{R}$. (5 '4 Marks)

Question 2

a. Define a norm on a vector space

b. Let X be a vector space. A function f :

$a \in [0,1]$. we have $f(ax + (1 - a)y) < af(x) + (1-a)f(y)$.

Verify the norm is a convex function.

c. Using 2(b), what can say about $\| \cdot \|$?

Question 3

Let $X = \mathbb{R}^2$ so that $x = (x_1, x_2)$. Sketch the unit sphere for norms on $\mathbb{R}^2$ given by (i) $\|x\|_1 = |x_1| + |x_2|$

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$\|x\|_2 = \sqrt{x_1^2 + x_2^2}$ (ii) $\|x\|_2 = \sqrt{x_1^2 + x_2^2}$ (iii) $\|x\|_\infty = \max\{|x_1|, |x_2|\}$ (6marks) |

b. Let P be the vector space of polynomials in t over $\mathbb{R}$ defined by

$P = \{a + 2bt : a, b \in \mathbb{R}\}$ such that $x(t) \in P$. Let $f, f_0: P \rightarrow \mathbb{R}$ be defined by

$f(x(t)) = \int x(t) dt$, $f_0(x(t)) = \int x(t) dt$. Find the basis of f, f_0 . (5marks)

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c. Consider the vectors (1) $Y = (1, -1, 0, 0, \dots)$ (11) $Z = (1, 1, 1, 1, \dots)$

(iii) $W = (-1, 1, -1, 1, \dots)$. Prove or disprove if (1) $\forall z, z \in \mathbb{N}$, (2) $y, Z, x \in \mathbb{N}$, (3) $\forall Z, X \in \mathbb{N}$.

(6 % marks)

Question 4 | Le)

q Let X be a vector space. Define Inner product on X and inner product space (3 marks)

b. Let $g(x) = (z, x)$ be defined. Prove that g is conjugate linear in the second factor (5marks)

c. State and prove Cauchy-Schwartz Inequality with respect to inner product space (9 Y% Mks)

Question 5

a. What is the dual of ℓ_p ? State the identity relation between them (2 marks)

b. Differentiate between a Banach space and a Hilbert space (5 marks)

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«Let $X = C[0,1]$ be endowed with the norm $\|f\| = \int_0^1 |f(t)| dt$ with $f(t) = 1-8t$,

$g(t) = 9t - 1$, compute (1) $\|g\|$ (2) $\|f\|$ (3) $\|f - g\|$ (10 marks)

Question 6 | ;) : h

q Let X be a usual norm, then define (i) Lipschitz map (11) contraction map (5 marks) A

| (5 marks)

». Prove that $T: \mathbb{R} \rightarrow \mathbb{R}$ defined by $T(x) = x + 3, x \in \mathbb{R}$ is a contraction.

e that if T is a linear map, then T is continuous at the origin imply that T is a Lipschitz map. .

eo ay (7 % marks)